

Why LLMs Remember First and Forget Last: Behavioral and Mechanistic Evidence

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Extension of: “[Transformers Remember First, Forget Last: Dual-Process Interference in LLMs](#)” · [PDF attached]

The Problem

Consider a match commentary feed a model is asked to track:

“Arteezy draws first blood — 1 kill at minute 3. ... A dominant midgame: Arteezy is now sitting at 8 kills, the lobby’s MVP. ... A clutch Roshan fight seals it — Arteezy closes out with 14 kills.”

Ask “How many kills did Arteezy have after first blood?” → models answer correctly ~70% of the time.

Ask “How many kills did Arteezy finish with?” → models answer correctly ~20% of the time.

The model reliably remembers the first recorded value but systematically fails to recall the most recent — the one that actually matters. We measure this as RI accuracy (recall first value, Retroactive Interference) vs PI accuracy (recall last value, Proactive Interference). The gap — RI–PI — is large, robust, and consistent. We call it primacy bias.

This is not a toy effect on a synthetic KV task. We confirm it on naturalistic Dota 2 match commentary (gap = +18pp, Qwen 1.5B) and on two independent KV datasets. The model is being asked a factual question about the state of the world right now, and it keeps reaching back to the beginning.

In our prior paper we documented this behaviorally across 39 models. This document describes the mechanistic extension: why does this happen?

Prior Work Already Tells Us What to Look For

Two recent lines of work establish the theoretical conditions under which this asymmetry must arise:

- Over-squashing in decoder-only Transformers [Barbero, Banino, Kumaran, Araújo, Veličković, Pascanu — NeurIPS 2024]: Proves that certain distinct input sequences yield arbitrarily close representations in the final token of a decoder-only Transformer. The last token’s representation is provably unable to distinguish them. The effect is exacerbated by low-precision floating point. This is the “Glasses” paper — the model needs glasses to see the end of the sequence.
- Attention sinks and anti-over-mixing [Barbero, Arroyo, Gu, Veličković, Bronstein, Pascanu — arXiv 2025]: LLMs attend heavily to the first token to avoid over-mixing of information. The first token acts as a sink that preserves early information while preventing later tokens from polluting it. Mechanistically, this is not a bug — it’s learned because it is useful.

Together these predict our finding exactly:

- First-token attention sink → first value is protected → RI is robust
- Last-token representation collapse (over-squashing) → most recent value degrades → PI fails

What was missing was a controlled behavioral measurement of the joint consequence, and mechanistic evidence of where in the network the collapse occurs. That is what this work provides.

What We Measured

11 models · 2 datasets · 9 key-levels × 12 update-levels · 100 trials/cell = 650 total test conditions

KV dataset: a controlled task where a key is assigned a value that is updated N times in sequence; the model is asked either for the first value (RI) or the most recent value (PI). Three variants tested:

- ARBITRARY_SINGLE — synthetic single-token English words (e.g. “cat”, “blue”); maximally controlled, no semantic priors; key - value might not be related
- SEMANTIC_MULTI — multi-token real-world values (e.g. names, locations) drawn from the same category; tests whether the effect survives richer content
- Dota 2 narratives — naturalistic match commentary generated programmatically from a Dota 2 game engine simulator; each “key” is a player, each “value update” is a kill-count change reported through realistic play-by-play text. Tests whether PI > RI survives when values are embedded in genuine narrative rather than a KV list (result: gap = +18pp, confirming the effect is not a format artifact)

Models span: Qwen2.5 (0.5B / 1.5B / 3B), Gemma-3 (270M / 1B / 4B), TinyLlama-1.1B, StableLM-1.6B, Pythia-

410M, Mamba-1.4B (SSM). Each model evaluated on a unified grid of 9 key-levels $\times$ 12 update-levels $\times$ 100 trials per cell — giving tight Wilson confidence intervals ($\pm 5\text{--}7\text{pp}$) and enough resolution to fit the exponential decay curve per model.

Model	RI accuracy	PI accuracy	Gap
Gemma-1B	57%	3%	+54pp
StableLM-1.6B	71%	5%	+66pp
Qwen 1.5B	63%	16%	+46pp
Mamba-1.4B (SSM)	70%	6%	+64pp
Gemma-4B	95%	49%	+46pp

Mean gap across instruction-tuned models: +43pp. PI accuracy follows exponential decay with updates N : $\text{PI}(N) = a \cdot e^{-bN} + c$, $R^2 = 0.83\text{--}0.97$.

Numbers above from ARBITRARY_SINGLE. Effect replicates on SEMANTIC_MULTI and Dota 2 narratives (see dataset descriptions above).

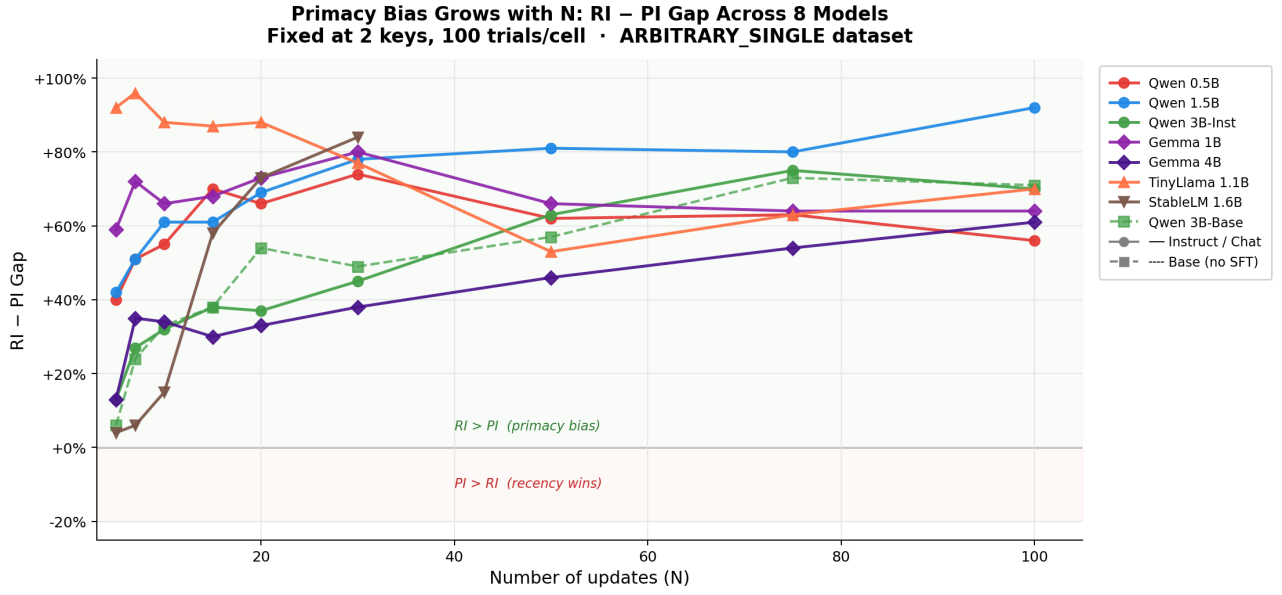


Figure 1: RI–PI gap vs number of updates N across 8 models (2 keys, 100 trials/cell). All instruction-tuned models stay above zero and the gap grows with N . Qwen 3B-Base (dashed) starts near zero, reflecting the base model effect. Solid lines = instruct/chat; dashed = base.

Mechanistic Analysis Suite

We ran a full interpretability pipeline across 6–8 models per experiment:

Experiment	Technique	What it asks
Logit lens	Layer-wise vocabulary projection	At which layer does v_{last} appear — and where does it get suppressed?
Linear probing	Probe on residual stream per layer	Does the model encode RI/PI correctness in its representations?
Attention attribution	Activation patching, head-level $\Delta P(v_{\text{last}})$	Which heads contribute to v_{last} retrieval vs suppression?
Causal ablation	Zero-out top-k heads, measure accuracy drop	Is the mechanism localised or distributed?
Jacobian at init	$\partial \text{output} / \partial \text{input}$ position norms, untrained model	Is the primacy bias present before any training?
Training dynamics	Full behavioral sweep across 40+ checkpoints	When during pretraining does the bias consolidate?
SAE feature analysis	Gemma Scope sparse autoencoder (Gemma-1B)	Do interpretable features activate selectively for v_{first} vs v_{last} ?

Three Mechanistic Findings

1. The bias is present before training — it’s in the architecture geometry

Jacobian analysis at random initialization (before any training) shows that position 0 influences the output $2.1\times$ more than middle positions (Qwen 0.5B). For Mamba SSM: $295\times$ (explained by HiPPO matrix transient

growth). The primacy bias is geometric — it exists in the untrained model’s computational structure, not in learned weights.

2. The correct answer is computed, then lost — logit lens locates where

Layer-by-layer logit lens shows $P(v_{\text{last}})$ (probability of the most recent value) rises to a peak of 0.03–0.24 at ~85% network depth, then is consistently outcompeted in the final layers by adjacent values. Final-layer $P(v_{\text{last}}) = 0.00\text{--}0.04$.

For v_{first} (RI): $P(v_{\text{first}})$ rises monotonically from early layers to 0.92–1.00 with no competition.

Logit Lens: The Model Finds the Last Value — Then the Final Layers Throw It Away

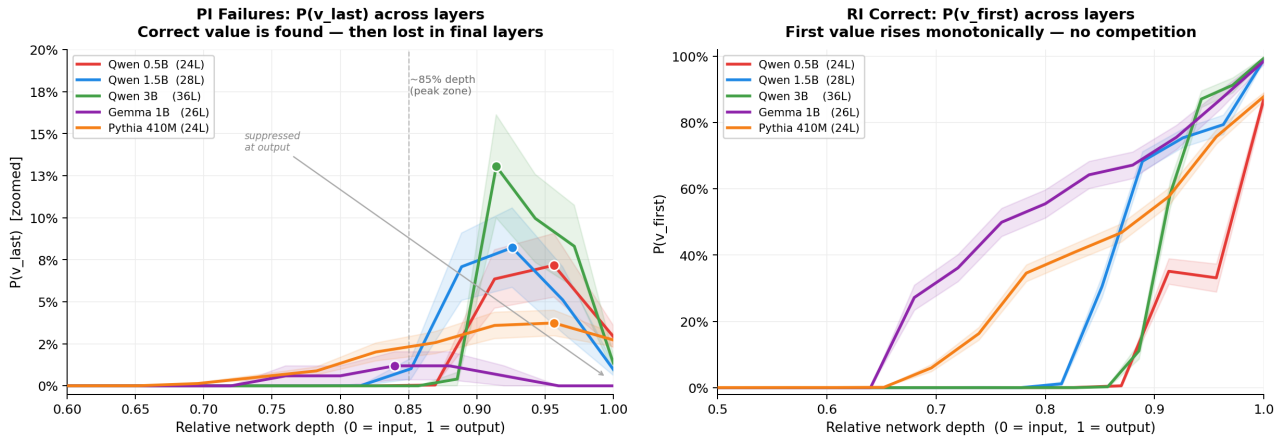


Figure 2: Left — PI failures: $P(v_{\text{last}})$ peaks at 81–92% network depth across 5 models, then collapses in the final layers (zoomed to the active region 0.6–1.0). Right — RI correct: $P(v_{\text{first}})$ rises monotonically to 0.87–1.00 with no suppression. The asymmetry is consistent across all architectures tested.

The model finds the right answer. The final layers throw it away. This is precisely the representational collapse Barbero et al. prove must occur.

3. The bias is fixed from the earliest pretraining steps — fine-tuning doesn’t cause it

Training dynamics on SmoLLM3-3B (full pretraining + SFT + alignment checkpoints):

- Step 40K (earliest checkpoint): gap = +59% — the bias is already present
- Gap remains positive through all pretraining stages, SFT, mid-training, and alignment
- SmoLLM2-1.7B: same pattern, positive from step 125K, persists to final model

Fine-tuning neither creates nor removes the primacy bias. It is established early in pretraining and is persistent.

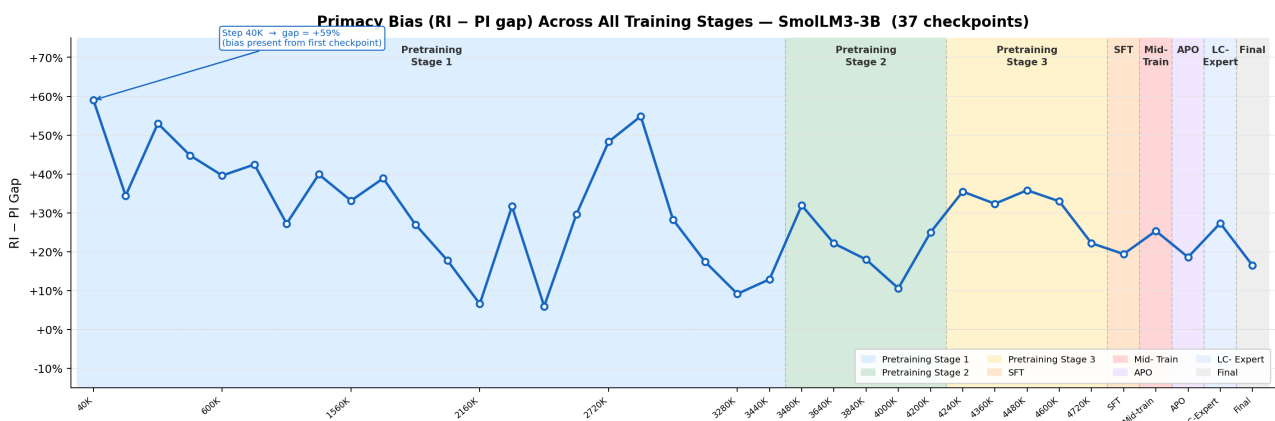


Figure 3: RI–PI gap across all training stages of SmoLLM3-3B (37 checkpoints). The bias is already +59% at step 40K — the very first checkpoint — and remains positive through all pretraining stages, SFT, mid-training, APO, and long-context fine-tuning.

What We Cannot Yet Explain

These are the open questions where collaboration would be most useful.

Q1 — Where is the suppression circuit?

Causal ablation shows no single bottleneck head responsible for suppressing v_{last} . Ablating the top contributing heads hurts retrieval (they help, not suppress). The mechanism is distributed across the final layer stack. What

is the architectural unit — attention pattern, residual stream interaction, MLP block — that implements the “forget the last value” behavior? Is there a graph-theoretic characterization of the computational graph that predicts suppression?

Q2 — Can we quantify the collapse magnitude from architecture parameters?

The Jacobian shows a $2.1\times$ primacy ratio for transformers but $295\times$ for Mamba at initialization. Is there a theory connecting d_{model} , sequence length, and number of layers to the interference ratio? The Glasses paper proves collapse is inevitable, but does not predict its magnitude for a given architecture. We would like to bound PI floor as a function of model capacity.

Q3 — Is there a phase transition in the bias during pretraining?

The gap is $+59\%$ at SmoLLM3 step 40K — essentially fully formed at $\sim 1\%$ through pretraining. Does the bias emerge from the first token processed, or is there an earlier phase where it is absent? What data or objective triggers consolidation of the primacy structure?

Q4 — What is the connection to over-squashing in the GNN sense?

Our logit lens shows the collapse occurs in the final 15% of layers. Is there an over-squashing interpretation — that the computational graph of the last few layers has too few paths from late-sequence tokens to the output? Or is it a competitive dynamics issue (multiple values competing for the same representation)?

Q5 — Can we fix it, and what does fixing it require architecturally?

We ran preliminary remedy experiments (landmark tokens, recency cues, numbered context) and found only marginal improvement — the bias resists surface-level prompt interventions. The deeper question is architectural: does fixing $PI > RI$ require changing the attention pattern (e.g. bidirectional encoding), the state capacity (e.g. larger d_{model} per position), or the positional geometry (e.g. replacing RoPE with a scheme that doesn’t amplify early positions)? Our bidirectional control (Flan-T5) shows no primacy bias — but at the cost of 64–77% garbage rate on the retrieval task, so bidirectionality alone is not sufficient. What is the minimal architectural change that preserves RI performance while restoring PI? This is the practical design question downstream of all the mechanistic findings.